

THE EFFECT OF *PANAX GINSENG* ON THE DEVELOPMENT OF TOLERANCE TO THE PHARMACOLOGICAL ACTIONS OF MORPHINE IN THE RAT

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(Received 6 September 1990)

Abstract—1. The effect of intraperitoneal administration of *Panax ginseng* on the development of tolerance to the analgesic and hyperthermic actions of morphine was determined in male Sprague–Dawley rats. Rats were rendered tolerant to morphine to different degrees by the subcutaneous implantation of either four pellets of morphine over a 3-day period or six pellets over a 7-day period. Each pellet contained 75 mg of morphine free base. Rats serving as controls were implanted with placebo pellets.

2. Daily administration of ginseng extract (6.25–50.0 mg/kg) for 3 days inhibited the development of tolerance to the analgesic effect but not to the hyperthermic effect of morphine in the four pellet schedule.

3. In six pellet schedule, daily administration of ginseng extract (25 and 50 mg/kg) for 7 days also inhibited the development of tolerance to the analgesic effect of morphine, but the 100 mg/kg dose had no effect. On the other hand, in six pellet schedule, the administration of ginseng extract (50 and 100 mg/kg) once daily for 7 days inhibited the development of tolerance to the hyperthermic effect of morphine.

4. It is concluded that in appropriate doses, ginseng extract has inhibitory activity on the development of tolerance to the pharmacological actions of morphine.

INTRODUCTION

The root of *Panax ginseng* has been used for medicinal purposes in the far Eastern countries for a long time. The chemistry and pharmacology of *Panax* has been reviewed recently (Shibata *et al.*, 1985). Ginseng has been shown to possess effects on the CNS and peripheral system. Among the actions studied are CNS depressant, anticonvulsant, antipsychotic and anti-fatigue actions (Shibata *et al.*, 1985). One of the active components of ginseng, ginsenoside R_{b1} was shown to suppress aggression in mice (Yoshimura *et al.*, 1988a, b).

Ginseng saponins appear to have analgesic activity (Nabata *et al.*, 1973; Ramarao and Bhargava, 1990) and hypothermic effect (Ramarao and Bhargava, 1990). Ginseng extract in high doses produced mild analgesic activity in the rat, which was not antagonized by naltrexone, an opiate agonist, suggesting that this effect was not mediated via opioid receptors or endogenous opioids (Ramarao and Bhargava, 1990). On the other hand, ginseng extract in low doses antagonized the analgesic, cataleptic and hyperthermic effects of morphine in rats (Ramarao and Bhargava, 1990). Kim *et al.* (1987) have demonstrated that ginseng can inhibit the development of tolerance and physical dependence on morphine in mice by multiple injections, although the mechanism of such action is not known.

The degree of inhibition of the tolerance to the pharmacological effects of morphine by various inhibitors depends on the dose and duration of the morphine treatment, as well as the inhibitor and method of induction of tolerance (Bhargava, 1988; Bhargava and Ramarao, 1989; Ramarao and Bhargava, 1990). For many years, research efforts in this laboratory have concentrated on finding drugs from natural or synthetic sources which could inhibit opiate tolerance–dependence process (Bhargava, 1986). The present studies were designed to determine if ginseng extract, which in low doses antagonizes various actions of morphine, can alter the development of tolerance to the pharmacological effects of morphine in rats implanted with the morphine pellets for different time periods.

MATERIALS AND METHODS

Animals

Male Sprague–Dawley rats weighing 225–250 g (Sasco King Animal Co., Oregon, WI) were acclimatized to a room with controlled ambient temperature ($23 \pm 1^\circ\text{C}$), humidity ($50 \pm 10\%$) and a 12 hr dark–light cycle (L0600–L1800). The animals were given food and water continuously.

Drugs

Standardized extract of *Panax ginseng* was a gift of Dr Hack-Seang Kim of Chongju, Korea. The ginseng extract was dissolved in distilled deionized water and injected intraperitoneally in a volume of 2 ml/kg body weight. Morphine sulfate (Mallinckrodt Chemical Co., St Louis, MO) was dissolved in saline and injected intraperitoneally in a volume of 1 ml/kg body weight. All solutions were prepared fresh on the day of their use. Morphine and placebo pellets were supplied by Dr Richard Hawks and

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Quantification of the analgesic response to morphine

The analgesic effect of morphine was measured by using a tail-flick apparatus as previously described (Bhargava, 1977, 1978; Bhargava and Gulati, 1990). The tail-flick latencies to thermal stimulation were determined before and at various times up to 240 min after an injection of morphine (8 mg/kg, i.p.). The basal latencies were found to be approx. 2 sec. A cut-off time of 10 sec was used to prevent damage to the tail. The basal response was subtracted from the effect induced by morphine. The area under the time response curve was calculated for each rat. The data are expressed as mean area $\pm$ SEM. Eight rats were used for each treatment group. The analgesic response in different groups was compared by using the analysis of variance followed by Scheffe's *S*-test. A value of $P < 0.05$ or less was considered to be significant.

Measurement of colonic temperature

The rats were injected with morphine and their colonic temperature was recorded prior to and at various time intervals for a period of 240 min. The colonic temperature was measured by using a Cole-Palmer digital telethermometer (Model No. 8502-20) and a thermistor probe inserted to a constant depth of 5 cm. The animals were lightly restrained and the temperature was recorded until a constant reading was obtained (Bhargava, 1981). The data were analyzed as indicated above for analgesia.

Effect of ginseng extract on tail-flick response and body temperature

The effect of ginseng extract *per se* on rat tail-flick reaction time and body temperature was studied. The doses of ginseng used were 50, 100 and 200 mg/kg. In order to determine whether the analgesia and the hypothermia induced by ginseng were mediated by opiate receptors, the effect of opiate antagonist, naltrexone on ginseng (200 mg/kg, i.p.) induced effects were determined. Naltrexone (1, 3 and 10 mg/kg) or its vehicle was injected immediately prior to the injection of ginseng extract (200 mg/kg). The analgesic and hypothermic effects were determined in the rats as described above.

Effect of ginseng extract on the development of tolerance to the analgesic effect of morphine

Induction and assessment of tolerance. Rats were rendered tolerant to and dependent on morphine by subcutaneous implantation of four pellets (each containing 75 mg of morphine free base), over a 3-day period as described previously (Bhargava, 1977, 1978, 1980; Bhargava and Gulati, 1990). Briefly, the back of the rat was shaved. On day 1, under light ether anesthesia at 10 a.m., one pellet of morphine was implanted after making a small incision in the lower back. The second pellet was implanted at 4 p.m. on the same day. On day 2, two more pellets were implanted at 4 p.m. All the pellets were removed 70 hr after the first pellet implantation. Control rats were implanted with placebo pellets. To determine the effects, ginseng extract in appropriate doses was injected daily for 3 days. All the pellets were removed on day 4. The tolerance to morphine was estimated by determining the analgesic and hyperthermic response to morphine (8 mg/kg, i.p.) in placebo and morphine pellet implanted rats 6 hr after the pellet removal.

Tolerance to morphine was also induced by implanting six morphine pellets during a 7-day period. One, two, three pellets were implanted on day 1, 3 and 5, respectively. The pellets were removed in the evening of day 7. Ginseng extract or its vehicle were injected once daily in appropriate doses on all the 7 days at 10:00 a.m. Analgesic activity was determined on day 8.

Table 1. Effect of intraperitoneal administration of ginseng extract on analgesic response

Treatment	Dose (mg/kg i.p.)	Analgesic response AUC _{0-240 min} (sec min), mean $\pm$ SEM (N)
Vehicle		0.94 $\pm$ 0.58 (8)
Morphine	8.0	134.23 $\pm$ 2.01 (8)*
Ginseng	50.0	6.41 $\pm$ 4.96 (8)
Ginseng	100.0	3.90 $\pm$ 2.39 (8)
Ginseng	200.0	24.43 $\pm$ 5.80 (16)†
Ginseng ^a + naltrexone	1.0	41.28 $\pm$ 9.88 (8)
Ginseng ^a + naltrexone	3.0	37.35 $\pm$ 6.30 (8)
Ginseng ^a + naltrexone	10.0	34.58 $\pm$ 3.30 (8)

^a 200 mg/kg, ip.

* $P < 0.0001$ vs vehicle group.

† $P < 0.05$ vs vehicle group.

RESULTS

Effect of ginseng on the tail-flick response and body temperature

Administration of ginseng extract *per se* produced analgesic activity. The AUC_{0-240 min} following the injection of vehicle, morphine (8 mg/kg, i.p.) and different doses of ginseng (200 mg/kg, i.p.) is shown in Table 1. Analysis of variance followed by Scheffe's *S*-test indicated that only 200 mg/kg of ginseng produced analgesic effect ($F_{3/28} = 3.44$; $P < 0.05$, Table 1). The analgesic effect induced by ginseng (200 mg/kg) in the rat was not antagonized by naltrexone in doses of 1–10 mg/kg (Table 1). Figure 1 shows the change in colonic temperature after the administration of various doses of ginseng up to 2 hr. A lower dose (50 mg/kg) did not produce statistically significant effects. The hypothermic response induced by ginseng (200 mg/kg, i.p.) was not antagonized by any of the doses of naltrexone used (data not presented).

Effect of chronic administration ginseng extract on morphine responses in rats implanted with placebo pellets

Chronic administration of morphine by pellet implantation resulted in the development of tolerance to the analgesic action of morphine. As shown in Fig. 2(A), implantation of six morphine pellets decreased the analgesic response to morphine (8 mg/kg) to 5% of that obtained in placebo pellet

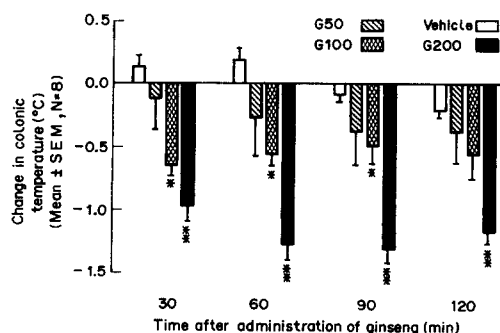


Fig. 1. Effect of ginseng extract (i.p.) on the colonic temperature of rats. The vertical lines within the bars represent the SEM values. G50, G100 and G200 are the ginseng 50, 100 and 200 mg/kg, i.p., respectively. * $P < 0.05$, ** $P < 0.001$ vs vehicle control.

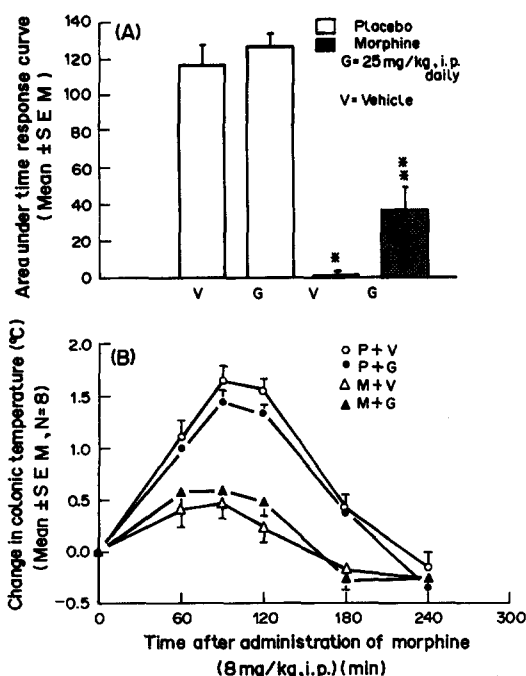


Fig. 2. Effect of ginseng extract (25 mg/kg per day) for 7 days on the development of tolerance to the analgesic (A) and hyperthermic effect (B) of morphine (8 mg/kg, i.p.). The rats were implanted with six morphine (M) or six placebo (P) pellets and injected with ginseng (G) extract or its vehicle as described in the text. * $P < 0.0005$ vs placebo pellets + vehicle; ** $P < 0.005$ vs morphine pellets + vehicle.

implanted rats ($F_{3/28} = 59.82$; $P < 0.0005$). Daily injection of ginseng extract (25 mg/kg) for 7 days did not modify the morphine induced analgesia in rats implanted with placebo pellets. However, the same treatment of ginseng produced a greater analgesic response to morphine in morphine pelleted rats. As evident from Fig. 2(B), ginseng (25 mg/kg, i.p. once daily for 7 days) did not alter morphine induced hyperthermic responses in rats implanted with placebo pellets. Implantation of morphine pellets resulted in the development of tolerance to the hyperthermic response of morphine which was not modified by daily injections of ginseng.

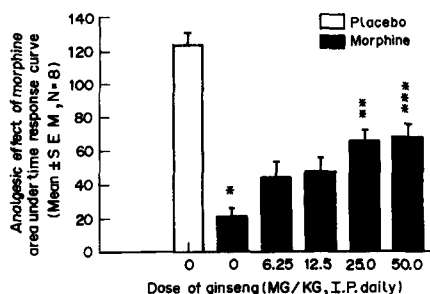


Fig. 3. Effect of ginseng extract on the development of tolerance to the analgesic effect of morphine. The rats were implanted with four morphine or four placebo pellets as described in the text and were injected with ginseng (6.25–50.00 mg/kg per day) for 3 days. * $P < 0.0005$ vs placebo treated group; ** $P < 0.0005$ vs morphine treated group; *** $P < 0.0001$ vs morphine treated group.

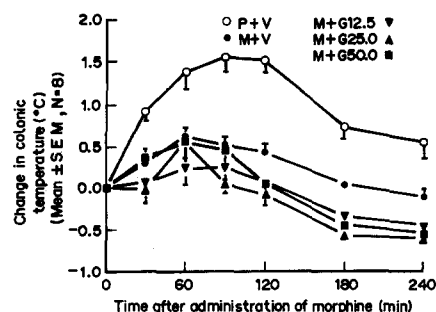


Fig. 4. Effect of ginseng extract (12.5–50.00 mg/kg per day) for 3 days on the development of tolerance to the hyperthermic effect of morphine. The rats were implanted with four morphine (M) or four placebo (P) pellets and injected with ginseng extract (G) as described in the text. The hyperthermic response to morphine (8 mg/kg) was determined in all treatment groups.

Effect of ginseng extract on the development of tolerance to the analgesic and hyperthermic effects of morphine

In subsequent experiments, the effect of different doses of ginseng on tolerance to morphine induced by implanting four or six morphine pellets was determined. Implantation of four morphine pellets resulted in the development of tolerance to morphine induced analgesia ($P < 0.05$). Daily injection of ginseng extract for 3 days in four pellet schedule dose dependently (6.25–50.0 mg/kg per day) inhibited the development of tolerance to the analgesic effect of morphine ($r = 0.89$, Fig. 3).

Implantation of four morphine pellets resulted in the development of tolerance to the hyperthermic action of morphine. ANOVA indicated a significant difference in the hyperthermic response to morphine pellet implanted rats. However, daily administration of ginseng extract (12.5–50 mg/kg) had no effect on the tolerance to hyperthermic action of morphine (Fig. 4).

The effect of ginseng extract on tolerance to morphine induced by implantation of six pellets during a 7-day period was also studied. This schedule induced a higher degree of tolerance to morphine than the four pellet implantation schedule ($F_{4/43} = 30.7$; $P < 0.0005$; Fig. 5) Daily injections of ginseng extract in 25 and 50 mg/kg doses inhibited tolerance development but 100 mg/kg dose produced lesser effect than 50 mg/kg dose.

The development of tolerance to the hyperthermic effect of morphine was inhibited by 50 and 100 mg/kg per day doses of ginseng administered daily for 7 days. Figure 6(A) shows changes in colonic temperature in response to morphine in various treatment groups. The same response converted into AUC is shown in Fig. 6(B).

DISCUSSION

The present studies indicate that chronic administration of morphine results in the development of tolerance to its analgesic effect in rats on the basis of results obtained using thermally evoked spinal reflex (e.g. tail-flick) and hyperthermic effect and that these

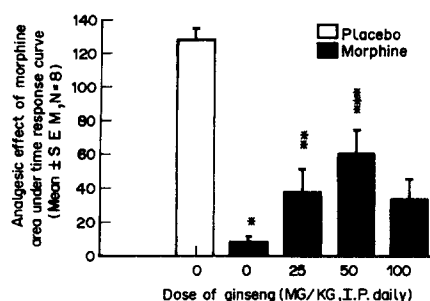


Fig. 5. Effect of ginseng extract (25–100 mg/kg per day) for 7 days on the development of tolerance to analgesic effect of morphine in rats implanted with six morphine pellets over a 7-day period. The analgesic response to morphine (8 mg/kg) was determined in all treatment groups. * $P < 0.0005$ vs placebo treated group; ** $P < 0.05$ vs morphine treated group; *** $P < 0.01$ vs morphine treated group.

processes are inhibited by daily administration of appropriate doses of ginseng extract.

In these studies, two different schedules of pellet implantation were used to produce tolerance to morphine. As expected the schedule utilizing six pellets during a 7-day period produced a greater degree of tolerance when compared to the schedule using four pellets during a 3-day period. In order to find inhibitors of opiate addiction processes, these initial studies, using the above mentioned morphine implan-

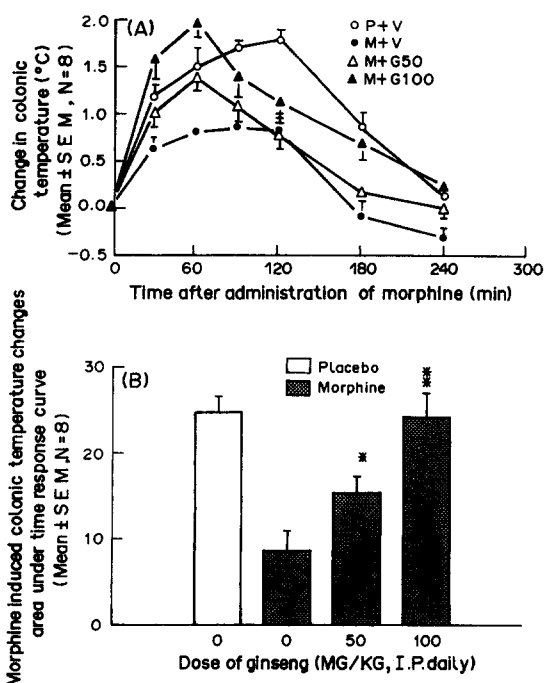


Fig. 6. Effect of ginseng extract (50 and 100 mg/kg per day) for 7 days on the tolerance to the hyperthermic effect of morphine (8 mg/kg, i.p.) in rats implanted with six morphine (M) or six placebo (P) pellets over a 7-day period. The time course (A) hyperthermic response to morphine was determined in vehicle (V) and ginseng (G) treatment groups. The latter was converted into area under the time response curve (B). * $P < 0.05$; *** $P < 0.001$ vs the vehicle injected morphine pelleted rats.

tation schedules, were conducted with varying doses of ginseng administered over several time periods. Daily injections of ginseng extract for 3 days in doses used inhibited the development of tolerance to the analgesic effect of morphine but not to its hyperthermic effect. However, long term treatment (seven daily injections) inhibited tolerance to both the analgesic and hyperthermic effects of morphine.

A recent study by Kim *et al.* (1987) has demonstrated that oral administration of ginseng saponins inhibited the development of tolerance to the analgesic action of morphine in mice. Tolerance was induced by injecting 10 mg/kg per day of morphine for 6 days and vehicle controls were not used.

Previous studies from this laboratory indicate that in the rat ginseng extract produces analgesia and hypothermia (Ramarao and Bhargava, 1990). Ginseng induced analgesia was not reversed by naloxone. Therefore, the inhibitory effect of ginseng on tolerance to morphine does not appear to be mediated via an action on opiate receptor systems or endogenous opioid systems.

Several chemical constituents of ginseng, mainly saponins, appear to have pharmacological activities. Our initial study was to determine whether ginseng has an inhibitory effect on tolerance to morphine. These initial studies will form the basis of further studies with pure components of ginseng and on the mechanism by which they inhibit the process by which tolerance to morphine is produced.

The observed decrease in the inhibition of development of tolerance to morphine analgesic response by ginseng extract (100 mg/kg daily for 7 days) needs some explanation. Similar phenomenon was reported by our laboratory with other inhibitors of tolerance to morphine such as cyclo(Leu-Gly) and MIF (Bhargava, 1988; Bhargava and Ramarao, 1989). At least two possible explanations for this phenomenon should be considered. First, ginseng extract on chronic administration (7 days treatment) with high dose can down-regulate the inhibitory system and can account for the loss of activity. However, this may not be true because chronic administration of ginseng extract for 7 days inhibited the development of tolerance to hyperthermic effect of morphine. The second and more plausible explanation would be that the decrease in the beneficial effect of ginseng extract on the development of tolerance to morphine analgesic-effect observed at high doses may result from an interaction of different doses of ginseng with different subgroups of receptors. It is possible that the ginseng extract used in the present study may be containing more than one active constituent having a different pharmacological profile. These observations indicate that the development of tolerance to the analgesic and hyperthermic effects of morphine involve different mechanisms.

In summary, the present studies show that ginseng extract can inhibit the development of tolerance to the analgesic and hyperthermic effects of morphine in the rat. Ginseng may be an important natural product because of its inhibitory action on the process of development of tolerance to morphine.

Acknowledgement—These studies were supported by a grant DA-02598 from the National Institute on Drug Abuse.

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